WordNet Similarity Measures for Legal Speech Analysis

Architecture and Methodology - Problem 15, MT-AIML (NLP Preprocessing & Language Modeling)

Twisha | B.Tech IT (AI & ML Honors), SCET, Sarvajanik University | Academic year 2026-27, Sem 6

1. Problem and scope

Legal speech-processing systems need a measure of conceptual closeness between the terms used in oral-argument transcripts and the terms defined in statutes. This project implements a semantic similarity tool over the WordNet noun hierarchy and compares two families of measures: **taxonomic** (Path, Wu-Palmer, Leacock-Chodorow), which use only the shape of the hierarchy, and **information-theoretic** (Resnik, Lin), which also use corpus statistics.

Scope: noun-to-noun similarity for single-word legal terms; one hand-checked WordNet sense per term; a small statute vocabulary (10 terms) for the transcript-to-statute matching demo.

2. WordNet taxonomy background

WordNet groups words into synsets (sets of synonyms with one gloss). Synsets are linked by hypernymy (is-a) into a hierarchy rooted at *entity.n.01* for nouns. A **lowest common subsumer (LCS)** of two synsets is their deepest shared hypernym. For example, contract.n.01 and agreement.n.01 are close in the hierarchy, while court.n.01 and judge.n.01 meet only at the root (entity.n.01).

3. Similarity formulas

Let len(a,b) be the shortest path (in edges) between synsets a and b, depth(c) the depth of synset c (root = 1), D the maximum depth of the noun taxonomy, and IC(c) = -log P(c) the information content of c, where P(c) is the corpus probability of c or any of its hyponyms.

Measure	Formula	Range	Idea
Path	1 / (len(a,b) + 1)	0 to 1	Shortest-path distance only
Wu-Palmer	2 * depth(LCS) / (depth(a) + depth(b))	0 to 1	Rewards deep LCS; depth-aware
Leacock-Chodorow	-log((len(a,b)+1) / (2D))	0 to ~3.64	Log-scaled path; D = 19 for nouns
Resnik	IC(LCS)	0 to max IC	Specificity of shared ancestor
Lin	2 * IC(LCS) / (IC(a) + IC(b))	0 to 1	Resnik normalised by both IC values

Relationship between Path and LCH. LCH is a monotonically decreasing function of path length, so for any set of pairs Path and LCH produce the *same ranking*. They differ only in scale (linear reciprocal vs logarithmic). This is verified empirically in the Evaluation Report.

Information content. Two IC tables are supported: the NLTK Genesis corpus (the suggested resource, with add-1 smoothing so unseen synsets do not receive near-infinite IC) and the Brown-corpus IC file shipped with NLTK. Genesis is small and not legal text, so its IC estimates are noisy; Brown is larger and more general. Both are reported.

4. System design and data flow

```
raw transcript (.txt)
   |  clean_text -> split_sentences -> tokenize -> lemmatize (WordNet morphy)
   v
```

```
legal lexicon (term -> synset)  --lookup-->  legal term occurrences
   |
   v
WordNetSimilarity.score(s1, s2) -> {path, wup, lch, res, lin}
   |-> pair table vs gold ratings   -> Spearman/Pearson, related-vs-control gap
   |-> term x term matrices         -> heatmaps
   |-> term vs statute definitions  -> best match + confidence threshold
   v
outputs/ (CSV, PNG, summary.json)  ->  Streamlit app, PDF reports
```

Modules. *preprocessing.py*: cleaning, sentence splitting (regex), tokenisation, lemmatisation, lexicon and data loaders. *models.py*: the five similarity measures, IC loading, matrices, correlation and gold-agreement analysis, statute matching. *pipeline.py*: evaluation runner. *app.py*: web interface.

Design decisions.

• One manually verified synset per term (data/annotated/legal_lexicon.csv). This isolates the similarity measures from word-sense-disambiguation errors, which are a separate problem (Module 4).

• Statute matching uses a confidence threshold on the score normalised to 0-1 (Path, Wu-Palmer and Lin are already bounded; LCH and Resnik are divided by the largest self-similarity in the candidate set). Below the threshold the tool reports 'no confident match' instead of forcing the nearest statute.

• Metrics are min-max normalised only for cross-metric comparison; raw scores are always kept.

• Sentence splitting is regex-based so the project runs without the punkt download.

5. Complexity

Scoring one pair requires finding hypernym paths for both synsets and their common ancestors, which is proportional to the taxonomy depth (at most 19 for nouns) times the number of senses' paths; IC lookup is O(1) from a precomputed table. A full n x n matrix over n terms therefore costs O(n^2) pair evaluations. For the 17-term matrices used here each metric takes roughly 0.1-0.2 seconds on the development machine.

6. Known limitations of the method

• Taxonomic measures treat every edge as equal distance, but WordNet is uneven: some regions are dense, others sparse. Equal path lengths therefore do not imply equal conceptual distance.

• WordNet is a general-English resource. Legal senses are present for many terms (e.g. tort, larceny, plaintiff) but legal relations such as 'judge presides over court' are not hypernym links, so *relatedness* (association) is only partly captured by *similarity* (is-a closeness).

• Information-content measures depend on the corpus. With add-1 smoothing the root synset *entity.n.01* receives a probability slightly above 1, giving it a small negative IC (about -0.083 for Genesis), so Resnik and Lin can be marginally negative for pairs whose only shared ancestor is the root.

• Only noun pairs are supported (LCH and Resnik require same part of speech).